Label-Free Crater Detection using Grounded SAM and Domain-Guided Self-Training

Dinis Corte Real
ISR/IST, LARSyS, Lisboa, Portugal
diniscortereal@tecnico.ulisboa.pt

Pedro M. Q. Aguiar
ISR/IST, LARSyS, Lisboa, Portugal
aguiar@isr.tecnico.ulisboa.pt

Margarida Silveira
ISR/IST, LARSyS, Lisboa, Portugal
msilveira@isr.tecnico.ulisboa.pt

***Abstract*—Impact craters provide essential data for planetary dating and mapping, yet their automated detection is hindered by morphological variability and high annotation costs. While supervised and semi-supervised deep learning methods achieve strong performance, they remain dependent on human-annotated craters for initialization or self-training, limiting scalability. We propose a fully label-free crater detection framework for Martian imagery that combines zero-shot foundation models with domain-guided self-training. Crater candidates are initialized via text-guided zero-shot detection using Grounded SAM and are subsequently refined through crater-specific geometric and photometric constraints, without any human annotations. To this end, we introduce an FBWR-enhanced DropLoss that leverages radial contrast symmetry to guide the iterative self-training procedure. Experiments on THEMIS near-infrared imagery show that our approach significantly improves upon zero-shot baselines and achieves approximately 86% of the AP50 of a fully supervised method, despite using zero labeled craters.**

***Index Terms*—crater detection, label-free learning, zero-shot detection, self-training, planetary remote sensing.**

I. Introduction

Impact craters are among the most informative geological structures on planetary surfaces, encoding evidence of impact flux, surface age, and geological evolution [1, 2]. On Mars, crater statistics underpin crater size–frequency analyses used for surface dating, geological mapping, and mission planning [3], while also providing stable landmarks for navigation and localization [4, 5]. As planetary missions continue to acquire high-resolution imagery at increasing scale, automated crater detection from remote sensing data has become a foundational requirement for planetary science and exploration [6, 7].

Manual crater cataloging is increasingly impractical due to the volume of data and the subjectivity introduced by crater degradation, illumination, and terrain variability [8, 9]. Consequently, automated crater detection has been widely studied, with supervised deep-learning approaches, particularly object detection frameworks, achieving strong performance on curated datasets [7, 10, 11]. However, these methods rely on extensive expert annotations and often generalize poorly across imaging conditions, sensors, and planetary domains [12, 13].

To reduce dependence on human annotations, prior work has explored weakly supervised, semi-supervised, and unsupervised learning strategies for object discovery and crater detection. In the computer vision literature, self-training and teacher-student frameworks enable object detection from unlabeled data through iterative pseudo-label refinement [14, 15]. However, these approaches either rely on generic objectness cues or require an initial set of human-annotated samples to anchor training. In crater detection, semi-supervised methods remain dependent on manually labeled craters for initialization or supervision [16, 17]. Consequently, existing crater detection pipelines have not eliminated the need for human-provided labels. Prior work has also shown that crater-specific domain knowledge, such as geometric and photometric constraints, can be incorporated into unsupervised or weakly supervised detection pipelines [18]. However, such priors are used independently of modern foundation models and have not been combined with zero-shot initialization and iterative self-training in a fully label-free setting, as we do in this paper.

Advances in self-supervised learning and vision foundation models have shown that object-centric representations can emerge without task-specific annotations, enabling zero-shot detection and segmentation guided by textual prompts alone [19, 20]. Naturally, these models have been applied in planetary science to identify crater-like structures across different bodies and sensing modalities without fine-tuning [21]. Nevertheless, zero-shot predictions typically yield limited recall, due to domain mismatch, scale variability, and crater degradation [22]. Moreover, remote sensing pipelines often reintroduce supervision through task-specific fine-tuning or use foundation models as fixed black-box predictors [23–25].

This paper introduces a fully label-free crater detection framework that integrates zero-shot initialization from a pre-trained foundation model with crater-specific domain constraints and iterative self-training. Initial pseudo-labels are generated using text-guided detection on unlabeled Martian imagery and refined through geometric and photometric criteria reflecting impact crater morphology [26, 27]. A domain-guided self-training strategy progressively refines pseudo-label quality without human intervention. Our experiments show that the performance of the proposed approach approximates that of a fully supervised method, without any supervision.

II. Methodology

A. Overview

The proposed crater detection approach combines zero-shot initialization from a foundation model with domain-guided

This work was partially supported by grants UNFORESEEN PROSSE/ESA (4000147138) and LARSyS FCT (10.54499/LA/P/0083/2020).

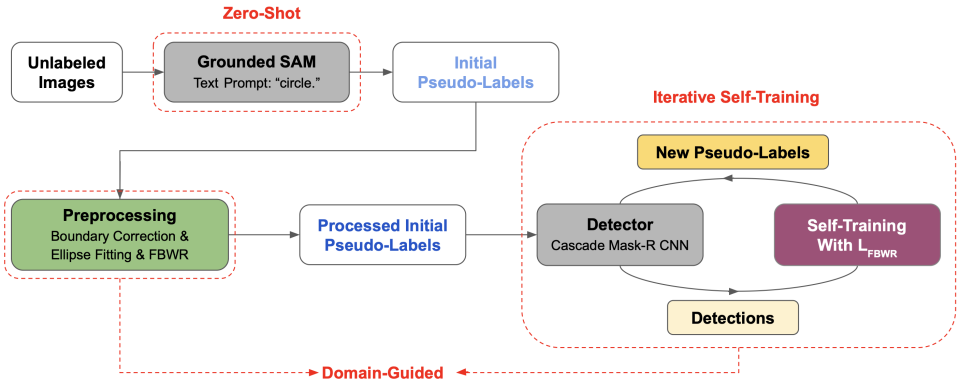


Fig. 1: Label-free crater detection pipeline.

pseudo-label refinement and iterative self-training, see Fig. 1.

Starting from raw images, initial crater candidates are generated using Grounded SAM via text-guided zero-shot detection. These candidates are refined using geometric and photometric constraints tailored to impact craters, yielding a higher-precision pseudo-label set. A Cascade Mask R-CNN detector [28] is then trained in an iterative self-training loop, where confident predictions are progressively incorporated as new pseudo-labels. Training is guided by a domain-aware loss function that mitigates pseudo-label incompleteness by encouraging the discovery of crater-like structures.

The final detector is trained on the stabilized set of pseudo-labels using the standard detection loss before inference, ensuring that the model learns accurate crater localization without being influenced by loss-dropping heuristics.

B. Data Description

Experiments are conducted on Martian surface imagery extracted from the global THEMIS daytime infrared (DIR) mosaic [29]. Non-overlapping image tiles of size 512×512 pixels are sampled across the mosaic, excluding regions above $\pm 50^\circ$ latitude to limit the geometric distortion introduced by cylindrical projection, as often done [30, 31]. The resulting data set has 9,122 image tiles, being split into 80% for training, 10% for validation, and 10% for evaluation.

Ground truth crater annotations are not used at any stage of training, pseudo-label generation, threshold selection, or self-training, serving exclusively for offline evaluation and analysis, using the Robbins and Hynek Mars crater catalog [8], following standard practice in unsupervised and zero-shot object detection.

C. Zero-Shot Pseudo-Label Initialization with Grounded SAM

Initial crater pseudo-labels are obtained using Grounded SAM [20], which combines text-guided object detection (Grounding DINO-Base [32]) with high-quality segmentation (SAM-Huge [19]). Naturally, prompt selection is relevant in this setting. Since the prompt *"crater"* yields low recall due to the limited representation in the language encoder's pretraining data, we adopt the prompt *"circle"*, which just exploits the approximately circular morphology of the craters. This choice produces broader and more diverse initial detections, which is advantageous for subsequent refinement. Detection thresholds are set conservatively (box threshold 0.3, text threshold 0.25) to prioritize precision at this stage.

The bounding boxes produced by Grounding DINO are fed to SAM, which uses default parameters. These bounding boxes, together with the segmentation masks provided by SAM, form the initial pseudo-labels, an incomplete and noisy set, but adequate for bootstrapping.

D. Domain-Guided Pseudo-Label Refinement

To suppress false positives and improve the initial pseudo-label's quality, we use a domain-guided refinement based on geometric and photometric crater priors. Geometric filtering removes detections with unrealistic eccentricity (axis ratio outside $[0.7, 1.3]$), while boundary correction adjusts truncated candidates near image borders by expanding the shorter bounding box dimension to recover circular crater shapes.

We further introduce the Flexible Black and White Rims (FBWR), a photometric criterion that validates crater-like rim shading patterns. In fact, impact craters typically exhibit asymmetric illumination, with opposing bright and dark rim segments [33]. FBWR formalizes this cue by defining a contrast- and gradient-weighted scoring function over diametrically opposed rim point pairs p, p':

$$\text{score}_{p,p'} = |I(p) - I(p')| \cdot \frac{\|\nabla I(p)\| + \|\nabla I(p')\|}{2},$$

where $I(\cdot)$ denotes pixel intensity and $\|\nabla I(\cdot)\|$ the gradient magnitude. The final FBWR score is obtained by averaging all valid point pairs across sampled centers and radii, see Fig 2. Detections with FBWR scores below a fixed and *a priori* defined threshold ($\tau^{\text{FBWR}} = 2000$) are discarded.

FBWR can be viewed as a variant of the Circle Hough Transform (CHT) [34]; however, unlike prior work that employed the CHT primarily for crater hypothesis generation [18], FBWR uses the CHT geometry to aggregate photometric evidence, embedding contrast and gradient information directly into the scoring function.

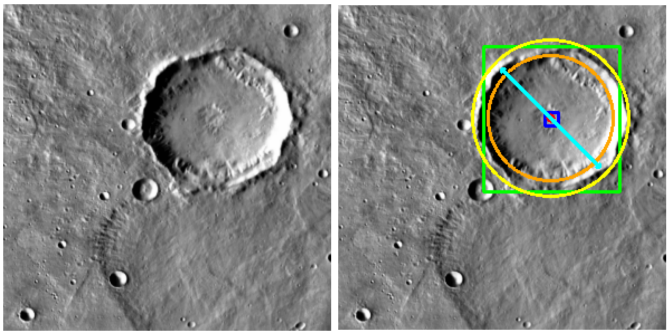


Fig. 2: Left: image. Right: corresponding analysis using FBWR. The bounding box (green) defines a central sampling region (dark blue) from which candidate centers (red) are selected. Radial vectors are evaluated between minimum (orange) and maximum (yellow) radii to assess rim consistency.

E. FBWR-Enhanced DropLoss for Self-Training

Training with pseudo-labels in a fully label-free setting is particularly challenging due to systematic missing annotations, as many true craters are absent from early pseudo-label sets. Penalizing predictions of such unlabeled objects can significantly hinder self-training. In the unsupervised object detection literature, CutLER [14] addresses this issue through DropLoss, a strategy that ignores predictions that do not overlap with any pseudo-label. By excluding these predictions from the loss, the detector is not penalized for discovering novel object instances during training. While effective for generic object discovery, CutLER relies solely on generic objectness cues and does not exploit domain-specific knowledge to distinguish plausible unlabeled objects from false positives.

Building on this insight, we extend DropLoss by incorporating domain knowledge through the FBWR score, yielding the proposed FBWR-Enhanced DropLoss. FBWR quantifies crater-like rim contrast independently of pseudo-label overlap. Predictions are excluded from the loss only if they both (i) have negligible overlap with existing pseudo-labels and (ii) exhibit strong FBWR responses, indicating plausible but currently unlabeled craters. Predictions with low FBWR scores are retained in the loss, even without overlap, as they are likely false positives.

Formally, for a predicted region r_i, the loss is defined as

$$\mathcal{L}(r_i) = \mathbb{1}\Big(\text{IoU}_i^{\max} > \tau^{\text{IoU}} \vee \text{FBWR}_i \leq \tau^{\text{FBWR}}\Big)\, \mathcal{L}_{\text{vanilla}}(r_i),$$

where $\text{IoU}_i^{\max}$ denotes the maximum IoU with the pseudo-label set and $\mathcal{L}_{\text{vanilla}}$ is the standard Cascade Mask R-CNN detection loss, The threshold τ^{FBWR} controls the exclusion of high-confidence crater candidates lacking pseudo-label support and is set to 0.03 to account for normalized image intensities (in $[0, 1]$) used during training, instead of raw 8-bit values (in $[0, 255]$).

F. Iterative Self-Training

The refined pseudo-labels are used to train a Cascade Mask R-CNN detector with the FBWR-Enhanced DropLoss. As in CutLER [14], we use multiple self-training rounds. After round t, predictions with confidence above $0.75 - 0.05t$ are added as pseudo-labels for round $t + 1$. Overlapping predictions (IoU > 0.5) replace existing annotations, enabling progressive expansion and refinement of the pseudo-label set. Our experiments have shown that three self-training rounds are sufficient to stabilize the performance. In fact, compared to generic self-training, our approach benefits from a stronger initialization and explicit crater-specific constraints, enabling robust learning and progressive refinement of crater detections.

III. Results and Discussion

A. Test Set Performance and Baseline Comparison

We evaluate the proposed approach against (1) the Grounded SAM baseline used as a simple zero-shot detector, and (2) an upper bound provided by a fully supervised Cascade Mask R-CNN. We report overall and crater size-specific performances, where crater subsets are defined based on bounding-box area: small (area $< \pi \cdot 5^2$), medium ($\pi \cdot 5^2 \leq$ area $< \pi \cdot 20^2$), and large (area $\geq \pi \cdot 20^2$). Although the pseudo-labels used for training consist of bounding-box and mask pairs, all evaluation metrics reported in this paper are computed on bounding boxes, since our focus is crater detection.

Table I reports the test set Average Precision (AP_{50}), where we can see that the proposed method substantially outperforms the zero-shot baseline and approximates the performance of the supervised detector, in particular for small and medium craters.

TABLE I: Test set Average Precision (AP_{50}). ZS, LF, and S denote zero-shot, label-free, and supervised, respectively.

Method	Type	All	Small	Medium	Large
Grounded SAM	ZS	9.8	1.0	20.4	37.4
Proposed Approach	LF	56.2	47.8	75.2	50.3
Cascade Mask-R CNN	S	65.0	53.3	80.9	72.0

Figure 3 plots Precision, Recall, F1 Score, and Accuracy as functions of IoU. Grounded SAM exhibits high precision but very low recall, reflecting its conservative zero-shot behavior. In contrast, the proposed approach achieves a more balanced trade-off, with recall approaching that of the supervised detector while maintaining competitive precision. The F1 Score and Accuracy of our method closely follow the ones of the supervised detector, confirming that iterative pseudo-label refinement improves coverage without compromising reliability.

These results demonstrate how foundation models, when combined with domain-guided self-training, can bridge a large portion of the performance gap between zero-shot and fully supervised crater detection methods. Figure 4 illustrates results obtained with the proposed method, where several particularly challenging small-sized craters are correctly detected.

B. Pseudo-Label Performance Study

1) Refinement of Zero-Shot Pseudo-Labels: We first evaluate the impact of domain-guided refinement on the initial zero-shot pseudo-labels. Table II shows that each refinement

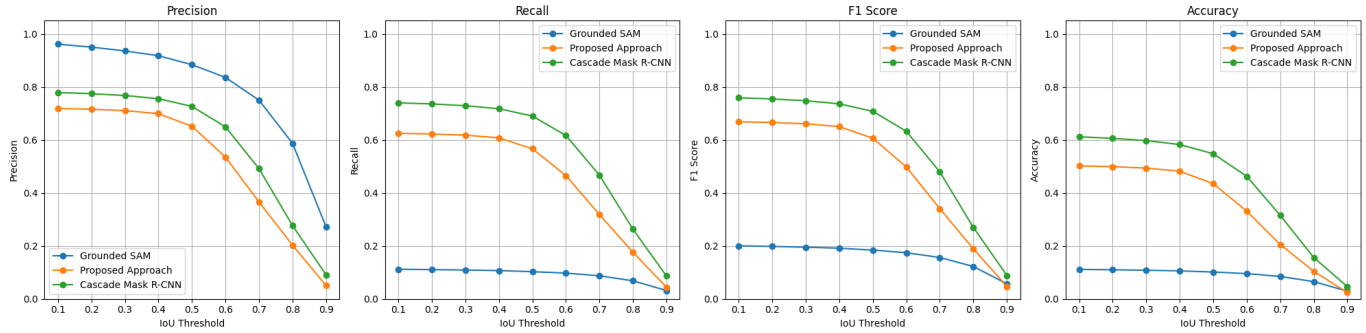


Fig. 3: Test set results for the Grounded SAM baseline, the label-free proposed method, and a fully supervised detector.

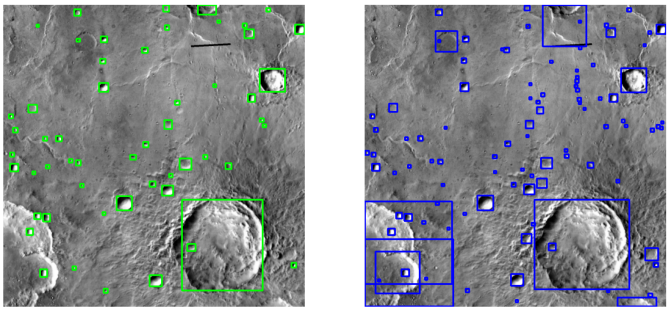


Fig. 4: Left: test set prediction examples. Right: ground truth.

step improves precision across IoU thresholds. Since the initial pseudo-labels represent only a small fraction of the true crater population, recall-oriented metrics are not informative at this stage. Instead, precision is the key indicator, as early false positives can irreversibly bias subsequent self-training.

Boundary correction yields the largest gain, indicating that many apparent false positives correspond to valid craters with misaligned bounding boxes near image borders. Ellipse fitting provides only marginal improvement, suggesting that most zero-shot detections exhibit near-circular geometry. FBWR further suppresses false positives by enforcing crater-specific photometric consistency, confirming the value of domain knowledge in refining zero-shot outputs.

TABLE II: Pseudo-label precision and recall at IoU = 0.5, for the initial and progressively refined pseudo-label sets, using boundary correction (BC), ellipse fitting (EF), and FBWR.

Pseudo-label Set	Precision	Recall
Initial	0.846	0.098
+ Boundary Correction (BC)	0.918	0.104
+ Ellipse Fitting (BC+EF)	0.926	0.097
+ FBWR (BC+EF+FBWR)	**0.951**	0.055

2) Effect of the Loss Function: To assess the impact of the proposed FBWR-Enhanced DropLoss, we compare its usage with that of the standard detection loss and the DropLoss [14], when starting from the same refined zero-shot pseudo-labels and undergoing three self-training rounds.

As shown in Table III, training with the standard loss yields high precision but does not significantly increase recall, indicating that the detector remains constrained by incomplete pseudo-labels. Both DropLoss variants enable recall to grow substantially across iterations, confirming the need for loss dropping for effective self-training. Compared to the original DropLoss, the proposed FBWR-enhanced variant achieves higher final precision at comparable recall, demonstrating that incorporating crater-specific photometric cues improves the reliability of newly discovered instances. This study confirms that domain-guided loss design plays a critical role in stabilizing label-free self-training.

TABLE III: Pseudo-label precision and recall at IoU = 0.5, across three self-training rounds, using the Standard Loss, DropLoss (DL), and the proposed FBWR-enhanced DropLoss (FBWR DL).

Method	Round 1	Round 2	Round 3
FBWR DL	0.868 / 0.274	0.733 / 0.452	0.638 / 0.503
DL	0.714 / 0.364	0.619 / 0.475	0.581 / 0.503
Standard Loss	0.947 / 0.094	0.950 / 0.098	0.951 / 0.100

IV. Conclusion

The paper introduces a label-free framework for Martian crater detection that combines zero-shot foundation models with domain-guided pseudo-label refinement and iterative self-training. Starting from text-guided detections produced by a pretrained vision–language model, the proposed pipeline progressively improves crater localization and coverage using geometric and photometric priors, most notably the Flexible Black and White Rims (FBWR) criterion, together with a domain-aware loss designed for incomplete supervision.

Experimental results show that the proposed approach substantially outperforms zero-shot baselines and recovers a large fraction of supervised performance without relying on manually annotated crater labels. Studies on pseudo-label performance confirm that both domain-guided refinement and the FBWR-Enhanced DropLoss play a critical role in stabilizing self-training and improving detection reliability, particularly for small and degraded craters.

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